

helixcode-status

HelixCode — Comprehensive Feature Status

Revision	8
Created	2026-06-19
Last modified	2026-06-19
Status	Active Development
Status summary	docs/features/helixcode-status_summary.md
Continuation	docs/CONTINUATION.md
Prior status	docs/features/Status.md (Revision 7, 2026-06-16)

Authoritative, in-depth inventory of **every** HelixCode feature across all services, infrastructure, client applications, LLM providers, and owned submodules. Kept in sync via the docs_chain engine (§11.4.106) and the Status-doc covenant (§11.4.45 / §11.4.53 / §11.4.56 / §11.4.57 / §11.4.153).

Anti-bluff (CONST-035 / §11.4.83 / §11.4.107): a feature is marked video-confirmed (Video: yes) ONLY when a real recorded scenario in /Volumes/T7/Downloads/Recordings shows it working end-to-end with a real LLM and that recording has been analyzed. No false “yes”. An un-recorded or un-analyzed feature is honestly Video: no / Video: pending, never bluffed green.

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Status dimensions (legend)

Column	Meaning	Values
Area	service / infrastructure / application / submodule	—
Component	package / tool / app / submodule	—
Feature	the discrete user-or-system capability	—
Dev	implementation status	done / partial / stub / absent
Wired	reachable from a shipped flow	yes / no / partial
Real-use	genuinely usable by an end user	yes / no / unknown
Tests	automated coverage	unit / integ / e2e / none (combinable)
V&V	captured runtime evidence (§11.4.5/ §11.4.69)	yes(path) / no
Video	recorded real scenario + analyzed (§11.4.83/ §11.4.107)	yes(path) / pending / no / n/a
Origin	native / ported:<cli_agent>	—
Overall	rollup	confirmed / working-untaped / partial / gap

Client Applications

App	Path	Dev	Wired	Real-use	Tests	Video	Overall
HTTP Server	cmd/server/	done	yes	yes	unit,integ	yes(helixcode-server-build-verified.mp4)	confirmed
CLI Client	cmd/cli/	done	yes	yes	unit,e2e	yes(helixcode-cli-demo-verified.mp4; helixcode-cli-generate-20260616.mp4; + 8 more)	confirmed
Terminal UI (TUI)	applications/terminal_ui/	done	yes	yes	unit,integ	yes(helixcode-tui-themed-20260615.mp4; helixcode-tui-llm-deepseek-20260616.mp4)	confirmed
Desktop (Fyne GUI)	applications/desktop/	done	yes	yes	unit,integ	yes(helixcode-desktop-chat-	confirmed

App	Path	Dev	Wired	Real-use	Tests	Video	O
Web Frontend	web/frontend/	done	yes	yes	e2e	themed-20260615.mp4; helixcode-desktop-themed-20260615.mp4) yes(helixcode-web-llm-console-deepseek-20260616.mp4; helixcode-web-themed-20260615.mp4)	co
Android (Kotlin)	applications/android/	partial	partial	yes	none	yes(helixcode-android-themed-20260615.mp4; helixcode-android-connect-20260615.mp4)	co
iOS (Swift)	applications/ios/	partial	partial	yes	none	yes(helixcode-ios-launch-20260615.mp4) [themed re-record OPERATOR-BLOCKED §11.4.52]	co
Aurora OS (Go/Fyne)	applications/aurora_os/	done	yes	yes	unit,integ	no	wo un
HarmonyOS (Go/Fyne)	applications/harmony_os/	done	yes	yes	unit,integ	no	wo un

Client application count: 8 (server, CLI, TUI, Desktop, Web, Android, iOS, Aurora OS, HarmonyOS)

Client Application Details

Server (cmd/server/): Gin HTTP server with REST API, WebSocket, pprof profiling, QA engine integration, LLMsVerifier integration. 56 API endpoints (12 public, 44 authenticated). Build: make build produces bin/helixcode.

CLI Client (cmd/cli/): Interactive REPL with 20+ commands, Cobra subcommand groups (permissions, worktree, hooks, mcp, commands, sessions, wizard, lsp, skills), real LLM generation via provider.Generate/GenerateStream, real shell execution via os/exec. BLUFF-001/002/003 all resolved.

Terminal UI (applications/terminal_ui/): tview-based TUI with chat streaming, env provider discovery, theme system, i18n, specify command, ensemble rendering, context usage display. 7 confirmed features with video evidence.

Desktop (applications/desktop/): Fyne GUI with chat streaming, theme system, i18n, nogui mode, brand/HXC theme, dashboard. Autonomous software-painter capture for LLM chat recording (no OS synthetic input needed).

Web Frontend (web/frontend/): Browser-based UI with LLM generate console, SSE streaming, specify phase form, response/metadata rendering. Viewport-scoped recordings at 1280x800.

Android (applications/android/): Kotlin app with RecyclerView task list, Go mobile-core bridge, Material dark theme. Models/settings/notifications UI is stub status.

iOS (applications/ios/): Swift app with UITableView task list, Go mobile-core bridge. Models/settings/notifications UI is stub status. Themed re-record OPERATOR-BLOCKED due to CoreSimulatorService write permission on /Volumes/T7.

Aurora OS (applications/aurora_os/): Full Go/Fyne multi-page GUI (dashboard/projects/sessions/tasks/LLM/workers/system), NoGUI CLI mode, theme system.

HarmonyOS (applications/harmony_os/): Go/Fyne 10-tab GUI with distributed engine, multi-device scheduling, NoGUI CLI, interactive shell, theme system.

LLM Providers

Provider	File	Has Tests	Live-Verified	Video	Status
OpenAI	openai_provider.go	Yes	Yes	—	Impl
Anthropic	anthropic_provider.go	Yes	Yes	—	Impl
Gemini	gemini_provider.go	Yes	Yes	—	Impl
Azure OpenAI	azure_provider.go	Yes	Yes	—	Impl
AWS Bedrock	bedrock_provider.go	Yes	Yes	—	Impl
Ollama	ollama_provider.go	Yes	Yes	—	Impl
LlamaCPP	llamacpp_provider.go	Yes	Yes	—	Impl
Mistral	mistral_provider.go	No	Yes	—	Impl
DeepSeek	deepseek_provider.go	No	Yes	—	Impl
Groq	groq_provider.go	Yes	Yes	—	Impl
xAI	xai_provider.go	Yes	Yes	—	Impl
OpenRouter	openrouter_provider.go	Yes	Yes	—	Impl
Qwen	qwen_provider.go	Yes (incl. OAuth)	Yes	—	Impl
KoboldAI	koboldai_provider.go	Yes	Yes	—	Impl
Copilot	copilot_provider.go	Yes	Yes	—	Impl
Vertex AI	vertexai_provider.go	Yes	Yes	—	Impl
Xiaomi MiMo	xiaomi_provider.go	Yes	Yes	yes(helixcode-xiaomi-api-verified.mp4; helixcode-xiaomi-integration-	Impl confi

Provider	File	Has Tests	Live-Verified	Video	Status
				verified.mp4; helixcode- xiaomi- challenge- verified.mp4)	
OpenAI Compatible	openai_compatible_provider.go	Yes	Yes	—	Impl
Ensemble	ensemble_provider.go	Yes	Yes	—	Impl (mul prov routi
Local	local_provider.go	Yes	Yes	—	Impl
Tool Provider	tool_provider.go	Yes	Yes	—	Impl

Provider count: 21 (19 with tests, 2 without: deepseek, mistral)

Provider Supporting Infrastructure

Component	File	Description
Auto LLM Manager	auto_llm_manager.go	Zero-touch provider initialization
Model Manager	model_manager.go	Model selection and management
Model Discovery	model_discovery.go	Dynamic model discovery
Load Balancer	load_balancer.go	Multi-provider load balancing
Health Monitor	health_monitor.go	Provider health tracking
Token Budget	token_budget.go	Token usage management
Cache Control	cache_control.go	Prompt caching
Usage Analytics	usage_analytics.go	Usage tracking
Provider Factory	provider_factory.go	Provider instantiation
Provider Registry	cross_provider_registry.go	Cross-provider model registry
Verifier Bridge	verifier_bridge.go	LLMsVerifier integration
Verifier Dynamic Catalogue	verifier_dynamic_catalogue.go	Dynamic model catalogue from verifier
Wizard	wizard.go	Interactive provider setup TUI

Component	File	Description
Aliases	aliases.go	Model alias management
Reasoning	reasoning.go	Reasoning/chain-of-thought support
Compression	compression/	Response compression
Prompt Cache	promptcache/	Prompt caching subsystem
Routing	routing/	Request routing
Vision	vision/	Vision/multimodal support
LiteLLM	litellm/	LiteLLM integration

Provider supporting components: 19

Internal Services

Inventory of every package under `helix_code/internal/` (67 packages). 64 have tests; 3 without (mocks, pprofutil, testutil) are test-support, not user features.

Package	Description	Has Tests	Wired	Status
adapters	Umbrella for consumer-side adapters (speckit_debate_adapter)	Yes	CLI	working-untaped
agent	Multi-agent orchestration and coordination	Yes	CLI	working-untaped
agentbridge	Wires HelixCode to real HelixAgent module	Yes	CLI	partial
approval	Central approval gate for user confirmations	Yes	CLI	working-untaped
approvalwire	CONST-046 message-ID resolver seam	Yes	CLI	working-untaped
auth	Authentication and authorization (JWT, bcrypt, sessions)	Yes	Server	working-untaped
autocommit	Auto-commit functionality for agent operations	Yes	CLI	working-untaped
cache	Multi-tier cache (config-gated L1/L2/L3)	Yes	Server	working-untaped
checkpoint		Yes	CLI	

Package	Description	Has Tests	Wired	Status
	Workspace checkpoints for session state			working-untaped
clarification	Clarification question generation for ambiguous prompts	Yes	CLI	working-untaped
clientcore	Shared agentic-capability wiring for all clients	Yes	CLI	partial
cognee	Integration with Cognee knowledge graph and memory	Yes	Server	working-untaped
commands	Slash command system with Markdown-based definitions	Yes	CLI	working-untaped
config	Comprehensive configuration management (Viper-based)	Yes	Server, CLI	working-untaped
context	Context building and management for AI conversations	Yes	CLI	working-untaped
continua	Continue.dev IDE integration	Yes	CLI	working-untaped
database	PostgreSQL connectivity and schema management (pgx)	Yes	Server	working-untaped
deployment	Deployment management	Yes	Server	working-untaped
discovery	Service/resource discovery (TTL, health, UDP multicast)	Yes	Server	working-untaped
editor	Editor integration (unified-diff, whole-file, search-replace)	Yes	CLI	working-untaped
ensembleui	Display-formatting helpers for ensemble provider rendering	Yes	CLI, TUI	partial
event	Event system (pub-sub, async/sync, task/workflow/worker types)	Yes	Server	working-untaped
fix	Fix/patch management	Yes	CLI	partial
focus	Focus tracking for sessions (hierarchical, priority-based)	Yes	CLI	partial
hardware		Yes	CLI	working-untaped

Package	Description	Has Tests	Wired	Status
	Hardware detection (CPU/GPU, optimal model-size inference)			
helixqa	HelixQA engine integration for autonomous QA sessions	Yes	Server	partial
hooks	Lifecycle hooks and event handling (priority, async/sync)	Yes	CLI	working-untaped
i18n_wiring	i18n wiring (legacy)	Yes	CLI	working-untaped
i18nwiring	Central boot-time CONST-046 translator wiring	Yes	CLI, Server	working-untaped
infraboot	Server infrastructure boot (containerized services)	Yes	Server	working-untaped
kilocode	AST-aware multi-file refactoring (call-graph, rename, refactor)	Yes	CLI	working-untaped
llm	Multi-provider LLM integration (21 providers)	Yes	Server, CLI	working-untaped
logging	Structured logging (logrus/zap, named loggers)	Yes	Server, CLI	working-untaped
logo	Image processing for brand asset management	Yes	CLI	partial
mcp	Model Context Protocol server (JSON-RPC 2.0, WebSocket, OAuth)	Yes	Server, CLI	working-untaped
memory	Long-term memory and conversation management	Yes	Server	working-untaped
mocks	Mock implementations for unit testing only	No	Tests only	n/a
monitoring	Metrics collection and system monitoring	Yes	Server	partial
notification	Multi-channel notification (Slack/Discord/Telegram/Email/Teams)	Yes	Server, CLI	working-untaped
performance	Production-grade performance optimization	Yes	Server	partial

Package	Description	Has Tests	Wired	Status
persistence	File-based state management (session serialization, auto-save)	Yes	CLI	working-untaped
planner	Task planner and step executor	Yes	CLI	partial
plantree	Plan tree system (branching, merging, node management)	Yes	CLI	working-untaped
plugins	Plugin loading from directory (base framework, activation)	Yes	CLI	partial
pprofutil	Opt-in pprof capture wiring	No	CLI	n/a
project	Project lifecycle management (DB-backed storage)	Yes	Server	working-untaped
projectmemory	Codex-style project-memory subsystem (loader, watcher, registry)	Yes	CLI	working-untaped
provider	Unified LLM provider interface definitions	Yes	Server, CLI	working-untaped
providers	AI and vector database integrations (fallback, load balancing)	Yes	Server	working-untaped
quality	Quality scoring/analysis (gate, build verification, linting)	Yes	CLI	working-untaped
redis	Redis client integration (go-redis, key-value, pub/sub)	Yes	Server	working-untaped
render	Fancy-mode terminal renderer (ANSI, frame buffer, streaming)	Yes	CLI	working-untaped
repomap	Semantic codebase mapping (symbol extraction, tree-sitter, ranking)	Yes	CLI	working-untaped
roocode	Roo-code CLI port (conversation store, code review)	Yes	CLI	partial
rules	Hierarchical project rule management (glob/regex/exact matching)	Yes	CLI	working-untaped
secrets		Yes	CLI	working-untaped

Package	Description	Has Tests	Wired	Status
	API-key loader (reads .env, api_keys.sh, missing-var detection)			
security	Comprehensive security management (scanning, zero-tolerance)	Yes	Server, CLI	working- untaped
server	HTTP server (Gin, JWT auth, 56 endpoints, QA handlers)	Yes	Server	working- untaped
session	Development session management (lifecycle, modes, focus chain)	Yes	Server, CLI	working- untaped
substrate	Shared parallel-dispatch substrate (Unit interface, queue scheduler)	Yes	CLI	partial
task	Distributed task management (checkpointing, dependencies, Redis)	Yes	Server	working- untaped
telemetry	OpenTelemetry instrumentation (agent/ LLM/tool spans)	Yes	CLI	working- untaped
template	Reusable template management (variable substitution, built-in lib)	Yes	CLI	working- untaped
testutil	Testing utilities	No	Tests only	n/a
theme	Built-in UI themes (detection, loading, customization)	Yes	CLI, TUI, Desktop	working- untaped
tools	Comprehensive tool ecosystem (shell, browser, fs, LSP, MCP, git)	Yes	CLI	working- untaped
verifier	LLMsVerifier integration (single source of truth, CONST-036)	Yes	Server, CLI	working- untaped
version	Build version information and metadata	Yes	CLI	working- untaped
voice	Speech-to-text voice input (Whisper API, local fallback)	Yes	CLI	partial
worker	Distributed worker pool (SSH, Raft consensus, cgroup isolation)	Yes	Server	working- untaped
workflow		Yes	Server	

Package	Description	Has Tests	Wired	Status
workspace	DAG-based workflow execution (planning/building/testing/refactoring)	Yes	CLI	working-untaped
	Container-based per-task workspace management (Docker runner)			partial

Total internal packages: 67 (64 with tests, 3 test-support: mocks, pprofutil, testutil)

CLI Commands

Top-Level Commands (cmd/cli/)

Command	Description	Status
generate [prompt]	Generate LLM response for a prompt	Implemented, live
version	Show platform version, providers count, build info	Implemented
server	Start the HTTP server	Implemented, live
start	Start the auto-LLM manager with monitoring	Implemented
auto	Run in auto mode with background processes	Implemented
chat	Interactive REPL chat session (default mode)	Implemented, live

Cobra Subcommand Groups

Group	Subcommands	Status
permissions	list, add, remove, check	Implemented, tested
worktree	list, enter, exit, remove	Implemented, tested
hooks	list, validate, test, enable, disable	Implemented, tested
mcp	add, remove, list, test, auth, logs	Implemented, tested
commands	list, show, run, reload	Implemented, tested
sessions	list, show, delete	Implemented, tested
wizard	(interactive/non-interactive provider setup)	Implemented, tested

Group	Subcommands	Status
lsp	status, list-servers, restart, stop	Implemented, tested
skills	list, show, invoke, reload	Implemented, tested

Interactive/REPL Commands

Command	Description	Status
handleGenerate	Real LLM generation via provider.Generate/GenerateStream	Implemented, live
handleListModels	Query all configured providers for models	Implemented, live
handleHealthCheck	Check provider health status	Implemented, live
handleListWorkers	List registered workers	Implemented
handleAddWorker	Add a new SSH worker	Implemented
handleNotification	Send notifications	Implemented
handleCommand	Execute shell commands via os/exec	Implemented, live
handleInteractive	Full interactive REPL session	Implemented, live
handleCheckpoint	Create/list/restore workspace checkpoints	Implemented
handleDiff	Show diff against a ref	Implemented
handleDebate	Run a debate on a topic	Implemented
handleSpecify	Run HelixSpecifier specification phase	Implemented
handleUndo	Undo last operation	Implemented
handleQARun	Start a QA session	Implemented
handleQAList	List QA sessions	Implemented
handleQAReport	Get QA session report	Implemented
handleQAScreenshot	Get QA session screenshot	Implemented
handleQACancel	Cancel a QA session	Implemented

Other cmd/ Tools

Tool	Path	Description	Status
helix_config	cmd/helix_config/	Interactive provider/credential config wizard	Implemented

Tool	Path	Description	Status
config_test	cmd/config_test/	Config + provider-credential validator	Implemented
il8n	cmd/il8n/	il8n bundle/ translator tooling	Implemented
infrastructure	cmd/infrastructure/	Container/ k8s/registry readiness checks	Implemented
performance_optimization	cmd/performance_optimization/	pprof profiling + bottleneck analysis	Implemented
security_test	cmd/security_test/	Security test harness (PoC execution)	Implemented
security_fix	cmd/security_fix/	Finding ingestion + policy-driven fix	Implemented
security_fix_standalone	cmd/security_fix_standalone/	Batch parallel security fix + audit-trail	Implemented
security_scan	cmd/security_scan/	AST/pattern code scan + leak detect + SARIF	Implemented (no tests)

API Endpoints

Public Endpoints (no auth) — 12 total

Method	Path	Description	Status
GET	/health	Health check (DB + Redis)	Implemented, live
GET	/api/v1/health	API-namespaced health check	Implemented, live
GET	/api/v1/server/info	Server info	Implemented
GET		System metrics	Implemented

Method	Path	Description	Status
	/api/v1/metrics		
GET	/api/v1/llm/providers	List LLM providers	Implemented
GET	/api/v1/llm/providers/:id	Get specific LLM provider	Implemented
GET	/api/v1/llm/models	List LLM models	Implemented
GET	/api/v1/memory/systems	List memory systems	Implemented
GET	/api/v1/memory/stats	Memory statistics	Implemented
GET	/ws	WebSocket connection	Implemented
GET	/static/*	Static file serving	Implemented
GET	/	Web frontend index	Implemented

Authenticated Endpoints — 44 total

Method	Path	Description	Status
POST	/api/v1/auth/register	User registration	Implemented
POST	/api/v1/auth/login	User login	Implemented
POST	/api/v1/auth/logout	User logout	Implemented
POST	/api/v1/auth/refresh	Refresh JWT token	Implemented
GET	/api/v1/users/me	Get current user	Implemented
PUT	/api/v1/users/me	Update current user	Implemented
DELETE	/api/v1/users/me	Delete current user	Implemented
GET	/api/v1/workers	List workers	Implemented
POST	/api/v1/workers	Create worker	Implemented
GET	/api/v1/workers/:id	Get worker	Implemented
PUT	/api/v1/workers/:id	Update worker	Implemented
DELETE	/api/v1/workers/:id	Delete worker	Implemented
POST	/api/v1/workers/:id/heartbeat	Worker heartbeat	Implemented
GET	/api/v1/workers/:id/metrics	Worker metrics	Implemented
GET	/api/v1/tasks	List tasks	Implemented
POST	/api/v1/tasks	Create task	Implemented

Method	Path	Description	Status
GET	/api/v1/tasks/:id	Get task	Implemented
PUT	/api/v1/tasks/:id	Update task	Implemented
DELETE	/api/v1/tasks/:id	Delete task	Implemented
POST	/api/v1/tasks/:id/assign	Assign task	Implemented
POST	/api/v1/tasks/:id/start	Start task	Implemented
POST	/api/v1/tasks/:id/complete	Complete task	Implemented
POST	/api/v1/tasks/:id/fail	Fail task	Implemented
POST	/api/v1/tasks/:id/checkpoint	Create task checkpoint	Implemented
GET	/api/v1/tasks/:id/checkpoints	Get task checkpoints	Implemented
POST	/api/v1/tasks/:id/retry	Retry task	Implemented
GET	/api/v1/projects	List projects	Implemented
POST	/api/v1/projects	Create project	Implemented
GET	/api/v1/projects/:id	Get project	Implemented
PUT	/api/v1/projects/:id	Update project	Implemented
DELETE	/api/v1/projects/:id	Delete project	Implemented
GET	/api/v1/projects/:id/sessions	Get project sessions	Implemented
POST	/api/v1/projects/:projectId/workflows/planning	Execute planning workflow	Implemented
POST	/api/v1/projects/:projectId/workflows/building	Execute building workflow	Implemented
POST	/api/v1/projects/:projectId/workflows/testing	Execute testing workflow	Implemented
POST	/api/v1/projects/:projectId/workflows/refactoring	Execute refactoring workflow	Implemented
GET	/api/v1/sessions	List sessions	Implemented
POST	/api/v1/sessions	Create session	Implemented
GET	/api/v1/sessions/:id	Get session	Implemented
PUT	/api/v1/sessions/:id	Update session	Implemented
DELETE	/api/v1/sessions/:id	Delete session	Implemented
GET	/api/v1/system/stats	System statistics	Implemented

Method	Path	Description	Status
GET	/api/v1/system/status	System status	Implemented
POST	/api/v1/llm/generate	LLM generation (real providers)	Implemented, live
POST	/api/v1/llm/stream	LLM streaming (real providers)	Implemented, live
POST	/api/v1/specify	HelixSpecifier specification phase	Implemented
POST	/api/v1/qa/session	Start QA session	Implemented
GET	/api/v1/qa/sessions	List QA sessions	Implemented
GET	/api/v1/qa/session/:id/status	QA session status	Implemented
GET	/api/v1/qa/session/:id/report	QA session report	Implemented
GET	/api/v1/qa/session/:id/screenshot/:name	QA screenshot	Implemented
DELETE	/api/v1/qa/session/:id	Cancel QA session	Implemented
GET	/api/v1/screenshot/engines	List screenshot engines	Implemented
GET	/api/v1/screenshot/:platform	Capture screenshot	Implemented

Total API endpoints: 56 (12 public, 44 authenticated)

Owned Submodules

Core Platform Submodules (HelixDevelopment)

Submodule	Path	Purpose	Integrated	Wired
HelixConstitution	constitution/	Governance rules, CLAUDE.md, AGENTS.md	Yes (cascaded)	yes
HelixQA	helix_qa/	QA/challenge orchestration platform	Yes (server)	yes
HelixAgent	submodules/helix_agent/	Agent framework	Yes (CLI)	yes
HelixLLM	submodules/helix_llm/	LLM orchestration	Yes	partial

Submodule	Path	Purpose	Integrated	Wired
HelixMemory	submodules/ helix_memory/	Memory subsystem	Yes	partial
HelixSpecifier	submodules/ helix_specifier/	Specification generation	Yes (CLI specify)	yes
LLMOrchestrator	submodules/ llm_orchestrator/	LLM orchestration	Yes	partial
LLMProvider	submodules/ llm_provider/	LLM provider abstraction	Yes	partial
LLMsVerifier	submodules/ llms_verifier/	Model verification (CONST-036)	Yes (server, CLI)	yes
VisionEngine	submodules/ vision_engine/	Vision/ multimodal processing	Yes	partial
DocProcessor	submodules/ doc_processor/	Document processing	Yes	partial
DebateOrchestrator	submodules/ debate_orchestrator/	Debate orchestration	Yes	yes
DagOrchestrator	submodules/ dag_orchestrator/	DAG-based orchestration	Yes	partial
PipelineRuntime	submodules/ pipeline_runtime/	Pipeline execution runtime	Yes	partial

Core Platform Submodules (vasic-digital)

Submodule	Path	Purpose	Integrated	Wired
Containers	containers/	Docker/ container artifacts (§11.4.76)	Yes	yes
Challenges	challenges/	Cross-cutting challenge bank	Yes	yes
Security	security/	Security tooling	Yes	partial
Panoptic	submodules/ panoptic/	Panoptic monitoring	Yes	no (out-of- process)
DocsChain	submodules/ docs_chain/	Documentation sync engine (§11.4.106)	Yes	yes

vasic-digital Library Submodules (48 total)

Agentic, Auth, AutoTemp, BackgroundTasks, Benchmark, Cache, Claritas, Concurrency, Config, Conversation, Database, Document, Embeddings, EventBus, Filesystem, Formatters, GandalfSolutions, HyperTune, I-LLM, I18n, Lazy,

LeakHub, LLMOps, MCP_Module, Memory, Messaging, Middleware, Models, Normalize, Observability, Optimization, Ouroborous, Planning, PliniusCommon, Plugins, RAG, RateLimiter, Recovery, RedTeam, SelfImprove, SkillRegistry, Storage, Streaming, ToolSchema, TOON, VectorDB, Veritas, Watcher.

Third-Party Dependency Submodules (3 total)

Submodule	Path	Purpose
llama.cpp	dependencies/ LLama_CPP/	Local LLM inference
Ollama	dependencies/Ollama/	Local LLM management
HuggingFace Hub	dependencies/ HuggingFace_Hub/	Model hub integration

CLI Agent References (55 total)

55 CLI agent reference implementations including: aider, claude-code, cline, codex, gemini-cli, plandex, qwen-code, crush, openhands, and many more.

Total submodules: 129

Video Confirmations

Feature Video Confirmations (§11.4.153)

Feature	Video	Content Verified	Date
Xiaomi API	helixcode-xiaomi-api-verified.mp4	Yes (4/4 patterns)	2026-06-15
Xiaomi Integration	helixcode-xiaomi-integration-verified.mp4	Yes (4/4 patterns)	2026-06-15
Xiaomi Challenge	helixcode-xiaomi-challenge-verified.mp4	Yes (4/4 patterns)	2026-06-15
Test Suite	helixcode-test-suite-verified.mp4	Yes (3/3 patterns)	2026-06-15
CLI Demo	helixcode-cli-demo-verified.mp4	Yes (4/4 patterns)	2026-06-15
Nano Clone	helixcode-nano-demo-verified.mp4	Yes (4/4 patterns)	2026-06-15
Server Build	helixcode-server-build-verified.mp4	Yes (2/2 patterns)	2026-06-15

Video-Confirmation Sweep 2026-06-16 (§11.4.153/§11.4.158)

Comprehensive real-DeepSeek video/recording confirmation sweep against fresh binaries. Every recording lives in /Volumes/T7/Downloads/Recordings with the helixcode- prefix, is a genuine real-LLM run, and has been analyzed.

CLI — 9 features (fresh bin/cli): - --stream (helixcode-cli-stream-20260616) - / generate (helixcode-cli-generate-20260616) - --list-models (helixcode-cli-list-models-20260616 — real deepseek-v4-flash/-pro catalog) - --command os/exec exit 0 (helixcode-cli-command-20260616) - --health (helixcode-cli-health-20260616) - --list-workers (helixcode-cli-list-workers-20260616 — 0 workers, honest) - --notify (helixcode-cli-notify-20260616 — dispatch runs, no enabled sink = honest caveat) - --model/-max-tokens cap (helixcode-cli-model-maxtokens-20260616) - --approval/-permission-mode (helixcode-cli-approval-mode-20260616)

API — 5 features (server :8080): - health (helixcode-api-health-20260616) - models — 7-model catalog (helixcode-api-models-20260616) - generate — real DeepSeek tokens=203 (helixcode-api-generate-20260616) - auth — register 201 / login→JWT / bad-pw rejected (helixcode-api-auth-20260616) - tasks-crud — 401 enforced + real UUID persisted (helixcode-api-tasks-crud-20260616)

TUI — 2 features (fresh bin/tui): - llm-chat DeepSeek “The capital of Japan is Tokyo.” (helixcode-tui-llm-deepseek-20260616) - navigation/theme tour (helixcode-tui-navigation-20260616)

Web — 2 features incl. SSE streaming (fresh server): - llm-console non-stream tokens=203 (helixcode-web-llm-console-deepseek-20260616.mp4) - SSE streaming “Python,JavaScript,Rust” (helixcode-web-04-deepseek-stream-20260616.png)

Themed Brand Re-Recordings (Rev 3, 2026-06-15)

5 of 6 clients have themed, real-LLM, window-scoped recordings: - helixcode-cli-themed-20260615.mp4 — brand banner + real DeepSeek “2+2 equals 4” - helixcode-web-themed-20260615.mp4 — dark theme + logo + real DeepSeek - helixcode-tui-themed-20260615.mp4 — brand tview.Styles + banner + real DeepSeek - helixcode-desktop-themed-20260615.mp4 — themed dashboard + embedded logo - helixcode-android-themed-20260615.mp4 — Material dark + logo + real connect

iOS themed re-record is OPERATOR-BLOCKED (§11.4.52):

CoreSimulatorService denied write to /Volumes/T7 for asset-catalog device-set creation.

Desktop Chat Re-Record (Rev 5, 2026-06-15)

helixcode-desktop-chat-themed-20260615.mp4 — autonomous Fyne software-painter capture (no OS synthetic input): real DeepSeek “2+2 equals 4” rendered in brand-themed pixels, 49 frames, 12.25s, liveness verified.

Test Evidence

Test Coverage Summary

Test Type	Status	Notes
Unit tests	All passing	64/67 packages have tests
Integration tests	All passing	Real PostgreSQL + Redis + LLM
E2E tests	All passing	Live API verification
Stress tests	All passing	Concurrent contention + sustained load
Chaos tests	All passing	Failure injection + recovery
Anti-bluff scan	Clean	No “simulated”/“for now”/“TODO implement”
§11.4.118/§11.4.135 guards	36 packages	Standing regression + -race guards

Anti-Bluff Payoff — The -stream §11.4.108 Stale-Binary Finding

The CLI `-stream` recording attempt initially produced invalid character `'d'` — a §11.4.108 STALE-BINARY break (the on-disk `bin/cli` predated a DeepSeek streaming-decode fix). Because the sweep READS the recorded screen, the break was caught instead of bluffed green. Fix: rebuilt `bin/cli` → streaming now emits real output; a §11.4.135 permanent regression guard was added (`internal/llm/deepseek_stream_guard_test.go`).

Coverage Completeness

Rev6 Gap-Pass (2026-06-16)

- **Internal packages:** 73/73 have ≥ 1 feature row. 234 internal-feature rows.
- **cmd tools:** all 11 `helix_code/cmd/*` dirs rowed.
- **Client apps:** all 8 surfaces (CLI, TUI, Web, Desktop, Android, iOS, Aurora OS, HarmonyOS).
- **HTTP API:** 18 endpoint groups rowed (56 endpoints).
- **LLM Providers:** 21 providers (19 with tests).
- **Submodules:** 50 inventoried (55 capability rows; principal features).
- **Ported cli_agents:** 33 rows (20 landed, 3 partial, 10 planned — honest).

Overall Rollup (from Status_Summary.md Rev 4)

Overall	Count	Meaning
working-untaped	324	real + tested, no analyzed video yet
partial	168	real but thin/unverified coverage
gap	49	scaffold / untested / planned
confirmed	23	real analyzed recording exists
n/a (test-support)	3	mocks/testutil/pprofutil
Total	567	

Inventory Sources

- helix_code/internal/ — 67 packages (72 directories including il8n, test dirs)
- helix_code/cmd/ — 11 command directories
- helix_code/applications/ — 8 client applications
- helix_code/internal/llm/ — 21 provider files + 19 supporting components
- helix_code/internal/server/ — 56 API endpoints
- docs/features/inventory/ — aggregated inventory sources
- /Volumes/T7/Downloads/Recordings/ — video evidence corpus
- docs/features/Status.md (Rev 7, 2026-06-16) — prior comprehensive status

Sources verified 2026-06-19

- Live API: All 56 endpoints verified via source code
- LLM Providers: 21 providers verified in helix_code/internal/llm/
- Client Applications: 8 apps verified in helix_code/applications/
- Internal Packages: 67 packages verified in helix_code/internal/
- Video Recordings: 30+ recordings in /Volumes/T7/Downloads/Recordings/
- Prior Status: docs/features/Status.md Revision 7 (2026-06-16)